

CONSTANTINE BOUDIAF INTL, Algeria

WMO: 604190

Lat: 36.276N

Lon: 6.620E

Elev: 2265

StdP: 13.53

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.6	33.4	26.4	22.2	40.8	28.5	24.4	40.4	26.7	48.2	23.8	48.1	3.2	300	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.5	102.4	68.1	98.9	68.0	95.5	67.6	72.4	91.5	71.0	89.8	69.7	88.3	11.2	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	109.3	78.2	65.8	103.6	77.6	64.2	97.9	76.6	37.5	91.8	36.2	89.5	35.0	87.9	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	22.3	19.0	16.6	26.5	107.8	3.5	1.9	24.0	109.1	21.9	110.2	19.9	111.3	17.4	112.6	
(5)				25.6	75.3	3.5	1.8	23.1	76.6	21.1	77.6	19.1	78.6	16.6	79.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.3	45.5	46.2	51.3	56.2	64.6	73.8	80.2	79.7	71.9	64.3	53.6	47.4
	DBStd	13.72	4.66	5.57	6.04	6.03	7.15	7.30	5.70	5.44	6.06	6.20	5.47	4.69
	HDD50	477	151	127	56	12	0	0	0	0	0	24	107	
	HDD65	2921	604	525	426	271	97	11	0	11	87	343	546	
	CDD50	4613	12	22	97	198	454	713	936	921	657	445	132	26
	CDD65	1581	0	0	1	7	86	274	471	456	218	67	1	0
	CDH74	18761	0	1	30	154	1115	3414	5989	5345	2014	676	22	1
	CDH80	9909	0	0	5	33	443	1799	3522	3014	866	226	1	0
Wind	WSAvg	5.8	6.3	6.7	6.2	6.1	5.7	5.8	5.7	5.4	5.1	5.0	5.9	5.8
Precipitation	PrecAvg	19.8	2.6	2.3	2.2	2.1	1.7	0.7	0.3	0.5	1.4	1.5	1.9	2.7
	PrecMax	34.2	9.1	7.1	6.2	7.6	4.9	3.6	1.9	1.6	4.3	5.6	5.6	12.9
	PrecMin	10.0	0.3	0.2	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.5	2.0	1.6	1.3	1.3	1.2	0.7	0.4	0.4	1.0	1.1	1.5	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.9	70.4	77.7	84.2	95.4	104.3	106.3	105.8	97.2	91.6	77.0	67.9
		MCWB	52.6	53.1	57.8	60.8	65.1	66.9	68.3	68.4	67.5	64.3	58.8	54.0
	2%	DB	61.1	64.7	71.8	78.5	89.5	99.1	103.0	101.9	93.1	86.0	71.7	62.6
		MCWB	50.5	52.4	55.7	60.1	64.3	66.7	68.7	68.2	67.4	62.9	56.2	51.8
	5%	DB	57.5	60.7	68.0	73.6	84.6	94.9	99.6	98.2	89.5	81.1	67.8	59.3
		MCWB	49.0	50.5	54.7	58.9	63.7	66.3	68.5	68.3	66.7	62.5	55.5	50.5
	10%	DB	54.4	57.1	64.2	69.5	80.4	90.3	95.9	94.6	84.6	76.9	64.1	56.3
		MCWB	47.3	48.6	52.9	56.9	62.7	65.9	68.1	67.9	65.7	61.1	54.1	49.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.3	55.8	60.2	64.6	69.8	72.3	74.5	74.5	72.2	68.5	61.7	55.3
		MCDB	62.7	65.3	72.6	77.6	85.7	92.0	95.8	92.7	87.4	81.8	72.2	64.8
	2%	WB	51.7	53.0	57.6	61.6	67.1	69.9	72.3	72.4	70.3	65.8	58.5	53.0
		MCDB	59.4	62.4	69.7	73.9	82.4	90.3	93.6	90.5	84.2	78.1	66.4	60.6
	5%	WB	49.6	51.1	55.4	59.5	65.1	68.2	70.8	71.0	68.6	63.9	56.6	51.3
		MCDB	56.4	59.8	65.9	71.0	80.5	87.6	92.1	89.0	82.3	76.0	64.6	58.1
	10%	WB	48.0	49.1	53.4	57.7	63.1	66.7	69.3	69.6	67.1	62.5	54.9	49.7
		MCDB	53.8	56.1	62.8	68.1	78.0	86.6	90.3	87.7	80.4	74.3	62.7	55.3
Mean Daily Temperature Range	5% DB	MDBR	16.8	17.9	20.3	21.8	24.9	27.4	28.5	27.1	23.1	21.6	18.2	16.7
		MCDBR	22.9	25.4	27.9	29.7	31.3	32.5	32.5	31.7	28.3	28.3	25.0	22.4
		MCWBR	14.5	14.9	15.2	15.4	13.0	11.5	11.6	11.2	10.1	11.6	13.1	13.7
	5% WB	MCDBR	21.3	23.0	25.7	26.1	27.9	28.8	28.6	26.9	24.3	23.8	21.6	20.0
MCWBR		14.0	14.1	14.8	14.6	13.0	11.8	12.2	10.9	9.8	11.0	12.5	12.8	
Clear-Sky Solar Irradiance	taub		0.308	0.328	0.384	0.420	0.451	0.440	0.481	0.461	0.438	0.391	0.345	0.315
	taud		2.483	2.421	2.239	2.158	2.093	2.150	2.026	2.082	2.165	2.278	2.400	2.479
	Ebn at Noon		282	289	279	274	266	268	256	259	258	260	264	271
	Edh at Noon		26	31	41	47	51	48	54	50	44	36	28	25
All-Sky Solar Radiation	RadAvg		767	1016	1403	1739	2037	2340	2359	2068	1585	1211	849	690
	RadStd		85	99	118	141	129	79	61	80	77	70	56	71

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	+0.70	+0.70	N/A	N/A	N/A	N/A

Nomenclature: See separate page